WEEK 1 · LESSON 2 OF 24 · HANDOUT

# Variables, Numbers and Text

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=kqtD5dpn9C8 — open the lesson online to watch inside the course.

**Your goal in this lesson:** Store values in variables, use the main data types (int, float, str, bool), do calculations, and build clear messages with f-strings.

## Variables are labelled boxes

A **variable** is a name that holds a value, so you can use it again.

```
shop_name = "Ineza Mini-Market"
milk_price = 800          # RWF per litre
litres_sold = 45

total = milk_price * litres_sold
print(total)
```

The `=` sign means **"store the value on the right in the name on the left"**. It is not "equals" like in maths.

**Naming rules:** use letters, numbers and `_`; start with a letter; no spaces. Choose clear names: `litres_sold` is better than `x`.

## The four basic types

| Type | Name | Example | Used for |
|---|---|---|---|
| `int` | integer | `45`, `-3` | whole numbers: counts, quantities |
| `float` | decimal number | `3.5`, `0.18` | prices with decimals, measurements, probabilities |
| `str` | string (text) | `"Huye"` | names, places, messages |
| `bool` | boolean | `True`, `False` | yes/no facts |

Check a type with `type()`:

```
print(type(45))
print(type(0.18))
print(type("Huye"))
print(type(True))
```

## Doing maths

```
price = 7500
vat_rate = 0.18                # Rwanda's standard VAT rate is 18%
vat = price * vat_rate
price_with_vat = price + vat
print(price_with_vat)          # 8850.0

print(17 // 5)   # whole-number division -> 3
print(17 % 5)    # remainder -> 2
print(round(3.14159, 2))   # 3.14
```

`//` and `%` are very useful: 17 eggs packed in trays of 5 gives 3 full trays and 2 eggs left over.

## Working with text

```
first = "Aline"
last = "Uwase"
full = first + " " + last      # joining strings
print(full)
print(full.upper())            # ALINE UWASE
print(len(full))               # number of characters: 11
print("  kigali ".strip().title())   # Kigali
```

## f-strings: the best way to build messages

Put `f` before the quotes and variables inside `{ }`:

```
customer = "Eric"
amount = 12500
print(f"Thank you {customer}, you paid {amount:,} RWF.")
```

`{amount:,}` adds thousands separators: `12,500`. Use `{value:.2f}` for two decimal places.

## Converting between types

Text that looks like a number is still text. Data often arrives as text, so you must convert it:

```
quantity_text = "12"
quantity = int(quantity_text)     # text -> int
price = float("850.5")            # text -> float
print(quantity * price)
print(str(2026) + " budget")      # number -> text
```

`"12" + 3` gives an error, because Python will not mix text and numbers without being told.

## Key terms

- **Variable** — a name that stores a value.
- **int / float** — whole numbers / decimal numbers.
- **str** — text, always inside quotes.
- **bool** — `True` or `False`.
- **f-string** — a string starting with `f` that can include variables in `{ }`.
- **Type conversion** — changing a value's type with `int()`, `float()` or `str()`.

**Remember this:** Variables store values; every value has a type. Do maths with numbers, join and clean text with string methods, and use f-strings for clear messages. Convert text to numbers before calculating.

Name: _____ Date: _____

## Activity

### Task 1

Create variables for a product name, its price in RWF and the quantity sold.

### Task 2

Calculate the total, the 18% VAT and the total with VAT.

### Task 3

Print one sentence with an f-string, for example: *"5 bags of Rice 5kg cost 37,500 RWF (44,250 RWF with VAT)."*

### Task 4

A customer types `" musanze "`. Clean it to `"Musanze"` with string methods.

## Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What does `=` do in `price = 800`?

   A. Deletes price
   B. Stores the value 800 in the variable named price
   C. Checks if price equals 800
   D. Prints 800

2. Which type is the value `0.18`?

   A. str
   B. int
   C. float
   D. bool

3. What does `17 % 5` give?

   A. 3
   B. 2
   C. 85
   D. 3.4

4. What does `f"Total: {amount:,} RWF"` do when amount is 12500?

   A. Gives an error
   B. Prints 'Total: 12,500 RWF'
   C. Prints 'Total: {amount:,} RWF'
   D. Prints 'Total: 12500.0 RWF'

5. Why does `"12" + 3` give an error?

   A. You cannot add text and a number without converting
   B. 12 is too big
   C. The quotes are wrong
   D. Python cannot add

6. Which is the best variable name?

   A. litres sold
   B. x
   C. 1litres
   D. litres_sold

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